MASTER IN AEROSPACE ENGINEERING

2MAE601 - Launchers Architecture

Rocket Launcher Design Based on Falcon 9 Baseline

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1 Introduction

This report focuses on designing a rocket launcher based on the Falcon 9 architecture. The mission involves inserting a payload of 23 tons into a Low Earth Orbit (LEO) with the following parameters:

- Perigee radius: $r_p = 250$ km
- Apogee radius: $r_a = 250$ km
- Orbit inclination: $i = 0°$

The launch site is located at latitude 0° and longitude −52.65°. A two-dimensional trajectory is assumed, with an azimuth angle $A = \pi/2$.

2 Computation of Δv requirements

The mission requirements form the foundation for the rocket design, specifying the velocity increments and losses needed to place the payload into the desired orbit. This section outlines the calculations required to achieve these requirements.
To achieve the mission, the total velocity increment Δv_{req} must be computed, accounting for both ideal and loss terms:

$$\Delta v_{req} = \sqrt{V_{orbit}^2 + V_0^2 - 2V_{orbit}V_0\cos(i)} + \Delta v_{losses} \tag{1}$$

where:

- $V_{orbit} = \sqrt{\frac{\mu}{R_E+r_p}}$ is the orbital velocity,
- $V_0 = R_E \cdot \omega$ is the ground velocity due to Earth's rotation,
- $\Delta v_{losses} = 1800$ m/s.

The total Δv combines the ideal orbital velocity and additional losses from gravity and drag. This computation ensures that the rocket is designed to meet the mission's velocity requirements.
Let's calculate each term. We need some constants: $R_{Earth} = 6371km$ (Earth radius), $M_{Earth} = 5.97601 * 10^{24}$ kg, G $= 6.670 * 10^{-11} m^3/kg/s^2$.

$$V_{\text{orbit}} = \sqrt{\frac{\mu}{(R_E + r_a)}} = 7759.018 \quad \text{m/s} \tag{2}$$

$$\omega = \frac{2\pi}{24 \cdot 3600} = 7.272 \cdot 10^{-5} \quad \text{rad/s} \tag{3}$$

$$V_0 = R_E \cdot \omega = 463.31 \quad \text{m/s} \tag{4}$$

$$\Delta v_{\text{req}} = V_{\text{orbit}} - V_0 + \Delta v_{\text{losses}} = 9095.706 \quad \text{m/s} \tag{5}$$

Equation 5 comes from equation 1 when i = 0.

3 Launcher Architecture and Masses

The launcher architecture defines the rocket's configuration, including the stages, propellant types, and structural indices. The mass calculations ensure proper balance between payload, propellant, and structural components.

3.1 First Stage and Second Stage data

The first stage is designed with 9 engines powered by LOX/RP1 propellants. This stage is responsible for providing the initial thrust needed to overcome Earth's gravity and atmospheric drag. The key properties are:

- Propellant densities: $\rho_{RP1} = 810$ kg/m^3, $\rho_{LOX} = 1140$ kg/m^3
- Specific impulse: $I_{sp,v,1} = 311.6$ s
- Structural index: $k_{s1} = 5.1\%$
- Pressure expansion ratio $\frac{p_c}{p_e} = 147.5$
- Chamber pressure $P_c = 125$ bar
- O/F = 2.3
- Efficiencies = $\eta_{c^*} = 0.97$, $\eta_{c_F} = 0.935$
- Throat area: $A_{t,1} = 0.042$ m^2
- Slenderness = L/D = 11.63
- Initial thrust-to-weight ratio: $T/W = 1.35$

The second stage is optimized for the vacuum environment and is responsible for finalizing the orbital insertion. Its properties are:

- Specific impulse: $I_{sp,v,2} = 348$ s
- Dry mass: $M_{s2} = 3.9$ tons
- Propellant mass: $M_{p2} = 92.67$ tons
- mass flow rate = 287.45 m/s
- Fairing mass: $M_{fairings} = 1.7$ tons
- Nozzle exit diameter: $D_{e,2} = 3.3$ m
- Slenderness = L/D = 3.45

In this section, we were asked to compute the Δv for each stage, the masses of each stage: dry and propellant and the GLOW (Gross Lift-Off Weight, the total weight of the rocket at launch).

3.2 Mass and Velocity Calculations

Let's start by defining the initial and final masses of each stage. For the first stage, we don't have all the necessary masses, whereas we have them for the second stage.

$$M_{i1} = M_{s1} + M_{p1} + M_{i2} + M_{\text{fairings}} \tag{6}$$

$$M_{f1} = M_{s1} + M_{i2} + M_{\text{fairings}} \tag{7}$$

$$M_{i2} = M_{s2} + M_{p2} + M_{Pl} = 119570.0\,\text{kg} \quad \text{(the fairings are jettisoned at stage separation)} \tag{8}$$

$$M_{f2} = M_{s2} + M_{Pl} = 26900.0\,\text{kg} \tag{9}$$

The total mass and Δv for each stage are calculated as follows. However, as we don't have enough data to calculate directly Δv_1, we need to calculate Δv_2 and then subtract it from the total one of the rocket, calculated in the previous section.

$$\Delta v_2 = g_0 I_{sp,v,2} \ln\left(\frac{M_{i2}}{M_{f2}}\right) = 5092.74 \quad \text{m/s}, \tag{10}$$

where M_{i2} and M_{f2} are the initial and final masses of the second stage, respectively, and $g_0 = 9.81 \quad m/s^2$.
Therefore, for the first stage:

$$\Delta v_1 = \Delta v_{\text{req}} - \Delta v_2 = 4002.96\,\text{m/s} \tag{11}$$

Concerning the masses, for the first stage, the propellant mass M_{p1} is computed using:

$$M_{p1} = \frac{(M_{f2} + M_{i2})(MR - 1)}{-MR + k_{s1} - MR \cdot k_{s1}} = 380438 \quad \text{kg}, \tag{12}$$

where:

$$\text{MR} = e^{-\frac{\Delta v_1}{g_0 \cdot I_{sp,\text{v},1}}} = \frac{M_{f1}}{M_{i1}} = 0.269 \tag{13}$$

The structural mass is:

$$M_{s1} = k_{s1} \cdot M_{p1} = 19402.3\,\text{kg} \tag{14}$$

It's important to notice that we can use $I_{sp,\text{v},1}$, and not the $I_{sp,\text{sl},1}$ one as we are taking account of the losses in the Δv_{losses}. Now, we have everything to calculate the initial mass of the rocket and the final mass of first stage:
equation 6: $M_{i1} = GLOW = M_{s1} + M_{p1} + M_{i2} + M_{fairings} = 521110\,\text{kg}$,
equation 7: $M_{f1} = M_{s1} + M_{i2} + M_{fairings} = 140672\,\text{kg}$.

There is another way to compute M_{i1} using the payload ratio λ. Such ratio is calculated as:

$$\lambda_1 = \frac{M_{pl1}}{M_i} = ((1 + k_{s1}) \cdot MR - k_{s1}) = 0.2327 \tag{15}$$

The combined mass of the second stage, payload, and fairing compose the payload "seen" by the first stage:

$$M_{pl1} = M_{s2} + M_{p2} + M_{pl} + M_{fairings} = 121270.0\text{kg} \tag{16}$$

The initial mass of the first stage is then computed as:

$$M_{i1} = GLOW = \frac{M_{pl1}}{\lambda_1} = 521110\text{kg} \tag{17}$$

This procedure yields the same result as the first method.

4 Propulsion Design

The propulsion design ensures that the rocket engines deliver the required thrust while maintaining efficiency. This section focuses on the engine performance parameters and their influence on stage design.
The propulsion system for the first stage includes 9 engines, and we are asked to calculate the following parameters:

- Thrust F_1 at sea level and in vacuum
- Exit nozzle area and diameter: A_{e1}, D_{e1}
- Mass flow rate: $\dot{m}_{1,\text{engine}}$
- Burning time: t_{b1}

Calculating the thrust at sea level of the complete first stage is straightforward:

$$F_{1sl} = M_{i1} \cdot g_0 \cdot T/W = 6901.30 \quad \text{kN} \tag{18}$$

To calculate the thrust in vacuum, we need to calculate other engine characteristics that are computed using the RocketCEA library (where fuel is RP1 and oxidizer is LOX in our case). Such engine characteristics are: the temperature of the chamber $T_c = 3683$ K, molar weight $M = 22.72$ g/mol and $\gamma = 1.149$. From these values, we can compute:

$$\Gamma = \sqrt{\gamma \cdot \left(\frac{2}{\gamma+1}\right)^{\frac{\gamma+1}{\gamma-1}}} = 0.638 \tag{19}$$

$$R = \frac{R_{\text{univ}}}{M} = \frac{8314.46\,\text{KJ/(kmol·K)}}{M} = \frac{8314.46}{22.72}\,\text{KJ/(kg·K)} = 365.84\,\text{KJ/(kg·K)}. \tag{20}$$

The characteristic velocity c^* is computed as:

$$c^* = \eta_{c^*} \cdot \frac{\sqrt{R \cdot T_c}}{\Gamma} = 1763.30\,\text{m/s}. \tag{21}$$

The expansion ratio ϵ is computed as:

$$\epsilon = \frac{A_e}{A_t} = \frac{\Gamma}{\sqrt{\left(\frac{2\gamma}{\gamma-1}\right) \cdot (p_c/p_e)^{-2/\gamma} \cdot \left(1 - (p_c/p_e)^{-(\gamma-1)/\gamma}\right)}} = 18.14. \tag{22}$$

The thrust coefficient C_F is calculated as ($p_{a,sl} = 101325$ Pa):

$$c_F = \left[\Gamma \cdot \sqrt{\left(\frac{2\gamma}{\gamma-1}\right) \cdot \left(1 - (p_c/p_e)^{-(\gamma-1)/\gamma}\right)} + \left(p_c/p_e^{-1} - \frac{p_{a,sl}}{p_c}\right) \cdot \epsilon\right] \cdot \eta_F = 1.60. \tag{23}$$

The specific impulse at sea level $I_{sp,\text{sl}}$ is computed as:

$$I_{sp,\text{sl},1} = \frac{c^* \cdot c_F}{g_0} = 286.78\,\text{s}. \tag{24}$$

The total mass flow rate $\dot{m}_1$ for one engine is:

$$\dot{m}_1 = \frac{F_{\text{sl}}}{g_0 \cdot I_{sp,\text{sl},1} \cdot 9} = 272.56\,\text{kg/s}. \tag{25}$$

The thrust in vacuum for one engine is ($\dot{m}_1$ is considered constant through the ascent):

$$F_{\text{vac,engine},1} = \dot{m}_1 \cdot I_{\text{sp,vac},1} \cdot g_0 = 833.17\,\text{kN}. \tag{26}$$

For all nine engines, the thrust in vacuum is:

$$F_{\text{vac,total},1} = 9 \cdot F_{\text{vac,engine},1} = 7498.57\,\text{kN}. \tag{27}$$

The exit nozzle area A_{e1} is:

$$A_{e1} = A_{t1} \cdot \epsilon = 0.76\,\mathrm{m}^2. \tag{28}$$

The nozzle diameter D_{e1} is:

$$D_{e1} = \sqrt{\frac{4 \cdot A_{e1}}{\pi}} = 0.98\,\mathrm{m}. \tag{29}$$

The burning time for 1 engine (and for all first stage as the engines will fire together) is:

$$t_{b1} = \frac{M_{p1}}{9 \cdot \dot{m}_{1,\mathrm{engine}}} = 155.09\,\mathrm{s}. \tag{30}$$

The burning time represents the time that one mass of propellant takes in order to provide a thrust equal to its weight. So, for the first stage, it has to be less than the specific impulse in order to provide a thrust higher than the weight. In other words, the one mass of propellant has to be burnt faster in order to provide more thrust than its weight. This inequality is respected for the first stage.
If we follow this reasoning, the burning time of the second stage should be higher than the specific impulse as, once in space, the thrust is mostly used for guidance and orbit insertion. In this case, however, as the thrust of the second stage is quite high, such inequality is not respected.

$$t_{b2} = \frac{M_{p2}}{\dot{m}_2} = 322.38 \quad \mathrm{s} \quad \prec \quad I_{sp,v,2} = 348 \quad \mathrm{s} \tag{31}$$

5 Geometry

The geometry of the rocket is critical for accommodating the propellant and payload while maintaining structural stability. In this section, we were asked to:

1. Estimate the dimensions of the two stages:
   - **First stage**: Determine its length and diameter.
   - **Second stage**: Determine its length and diameter.
   - **Fairing**: Calculate its length and diameter.

   Data and Constraints:

   - The diameter of the second stage is equal to the diameter of the first stage.
   - The diameter of the fairing is equal to the diameter of the first stage.
   - The height of the fairing is defined as:

     $$L_f = 3.614 \cdot D$$

   - For the first stage, the total volume is calculated as:

     $$V_1 = 1.06 \cdot V_p$$

     where V_p is the propellant volume, computed using the average density of the propellant.

2. Estimate the launcher initial reference area, defined as the launcher cross-sectional area at take-off.

Let's start from the first point. The volumetric density of the propellant ($\rho_{\text{volumetric}}$) is a measure of how much mass of the combined fuel and oxidizer can be packed into a given volume. It is a critical parameter in rocket design, as it directly affects the size of the propellant tanks and the overall geometry of the rocket.
It is calculated as:

$$\rho_{\text{volumetric}} = \frac{1 + OF}{\left(\frac{1}{\rho_{\text{RP1}}} + \frac{OF}{\rho_{\text{LOX}}}\right)} = 1016.72\,\text{kg/m}^3 \tag{32}$$

The volume of the propellant is:

$$V_{p1} = \frac{M_{p1}}{\rho_{\text{volumetric}}} = 374.18\,\text{m}^3 \tag{33}$$

The total volume of the first stage is:

$$V_1 = 1.06 \cdot V_{p1} = 396.63\,\text{m}^3 \tag{34}$$

To estimate the diameter of the first stage, I thought about the engines configuration on the Falcon 9 and I imagined that, to fit the engines, the diameter has to be at least 3 times the diameter of each engine, plus some additional space. In other words: $D_1 = D_{e1} \cdot 3 + i$. To calculate the correct i, I used an optimization cycle:

```
for i in np.linspace(0.4, 0.8, num=41):
    D1 = De1 * 3 + i
    L1 = slenderness_1 * D1
    V_try = np.pi*(D1/2)**2 * L1

    if np.abs(V_try - Vp1*1.06) < 0.1:
        break
else:
    print("No suitable value found.")
```

Such cycle guesses the diameter, calculates the length of the first stage using the given slenderness and then calculates the volume with these data. Such volume should be equal to total volume of the first stage V_1 calculated before. I took an error between the esti and the old volume of 0.1. Such cycle yields: diameter of first stage D_1 =3.514 m, length of first stage L_1 = 40.87 m.
We can also calculate the length of the tank of the first stage:

$$h_1 = \frac{V_{p1}}{\pi \cdot \left(\frac{D_1}{2}\right)^2} = 38.56\,\text{m} \tag{35}$$

The length of the second stage is:

$$L_2 = \text{slenderness}_2 \cdot D_1 = 12.13\,\text{m} \tag{36}$$

The length of the fairing is:

$$L_{fairings} = 3.614 \cdot D_1 = 12.70\,\text{m} \tag{37}$$

The total length of the rocket is:

$$L_{\text{total}} = L_1 + L_2 + L_{fairings} = 65.71\,\text{m} \tag{38}$$

For reference, the real Falcon 9 measures: 69.8 m in length and 3.7 m in diameter.
The launcher initial reference area is:

$$A_{\text{ref,F9}} = \left(\frac{D_1}{2}\right)^2 \cdot \pi = 9.70\,\text{m}^2 \tag{39}$$

6 Trajectory Optimization

The trajectory optimization verifies the rocket's ability to reach the desired orbit while minimizing losses. This section outlines the integration of the trajectory in the rotating reference frame.

6.1 Variables and specifications

The control and design variables are the following:

- mass of the payload (1 variable),
- delta pitch angle for the pitch over maneuver (1 variable),
- bilinear tangent law parameters for the second stage (3 variables).

The constraints are the following:

$$\begin{aligned} h(t = t_f) &= h_{\text{target}} \quad \text{(tolerance of 1 km)} \\ v(t = t_f) &= v_{\text{target}} \quad \text{(tolerance of 1 m/s)} \\ \gamma(t = t_f) &= \gamma_{\text{target}} \quad \text{(tolerance of 0.01 deg)} \end{aligned}$$

Several trajectory parameters are considered as fixed for the trajectory optimization:

- duration of vertical phase,
- duration of pitch-over maneuver phase,
- duration of exponential decay (second phase of pitch-over maneuver).

For this optimazation, a rotating Earth is considered.
The trajectory is defined as follows:

- **Vertical phase** with the pitch angle $\theta = 90$ deg.
- **A pitch-over maneuver** with two phases:
  - The evolution of the pitch angle (θ) to have a linear evolution of angle of attack ($\alpha = \theta - \gamma$), with respect to the following law:

    $$\theta = \gamma - \Delta_{\theta_{po}} \frac{t}{\Delta t_{po}}$$

    where γ is the flight path angle.
  - Exponential decay of angle of attack (α) leading to the evolution of the pitch angle:

    $$\theta = \gamma - \Delta_{\theta_{po}} \exp\left(-\frac{t}{3\Delta t_{\text{decay}}}\right).$$
- **A gravity turn phase**, in which the pitch angle is defined by $\theta = \gamma$ (i.e., $\alpha = 0$).
- **A bilinear tangent law of pitch angle for exoatmospheric flight** (flight of second stage):

  $$\theta = \arctan\left(\frac{a^\xi \tan(\theta_i) + (\tan(\theta_f) - a^\xi \tan(\theta_i))t'}{1 - a^\xi t'}\right),$$

  with t' the normalized time within the phase, θ_i and θ_f the initial and final values of pitch angle for this phase, a is fixed to 100, and ξ has to be optimized.

6.2 Velocity in the rotating frame

The trajectory is integrated in a rotating frame with the velocity computed as:

$$\vec{V}_{abs} = \vec{V}_{rel} + \vec{\Omega} \wedge \vec{OM}, \tag{40}$$

where $\vec{V}_{rel}$ is the relative velocity. The trajectory is computed in the rotating frame, so we need to compute:

$$V_{\text{rel}} = |\vec{V}_{\text{rel}}| \tag{41}$$

With the hypothesis of a launch through the East from a latitude of 0°, we can obtain:

$$V_{\text{rel}} = V_{\text{cin}} - V_{\text{rot}} \tag{42}$$

The rotating velocity at the injection point is:

$$V_{rot} = \omega \cdot (R_E + r_p) \cdot \cos(\text{lat}) = 481.49 \quad \text{m/s} \tag{43}$$

where $\omega = 7.272 \cdot 10^{-5}$ rad/s and lat = 0°. Equation 42 is thus: $V_{rel} = 7277.52$ m/s. With this value, we can now optimize the trajectory thanks to the *Trajectory-optimization* file.

6.3 Computation of losses

The variables used for the optimization are M_P (propellant mass), M_s (structural mass), $I_{sp,v}$ (specific impulse in vacuum), $\dot{m}$ (mass flow rate), $A_{\text{ref,F9}}$ (reference area), and A_{e} (nozzle exit area) for both first and second stage. Once the optimization is run and terminated successfully, the optimized payload mass is: 23.42 t, that is higher than the 23 t specified at the beginning. This means that we have overestimated the losses (=1800 m/s at the beginning). Indeed, the required Δ of velocity is:

$$\Delta v_{\text{req}} = \Delta v_{\text{propulsive}} - \text{loss}_{\text{nozzle}} - \text{loss}_{\text{AoA+gimb}} - \text{loss}_{\text{aero}} - \text{loss}_{\text{gravity}} = \Delta v_{\text{propulsive}} - \Delta v_{\text{losses}} \tag{44}$$

That can also be written as:

$$\begin{aligned} \int_{t_0}^{t_f} \dot{V}\, dt = \int_{t_0}^{t_f} \frac{T_{\text{vacuum}}}{m}\, dt - \int_{t_0}^{t_f} \cos(\theta - \gamma + \delta) \cdot \frac{P_a(r) A_n}{m}\, dt - \\ - \int_{t_0}^{t_f} \frac{D}{m}\, dt - \int_{t_0}^{t_f} g(r) \sin\gamma\, dt - \\ - \int_{t_0}^{t_f} (1 - \cos(\theta - \gamma + \delta)) \frac{T_{\text{vacuum}}}{m}\, dt \end{aligned} \tag{45}$$

where:

- $\dot{V}$: The change in velocity over time.
- T_{vacuum}: Thrust in vacuum conditions.
- m: The instantaneous mass of the rocket.
- $P_a(r)$: Atmospheric pressure as a function of altitude r.
- A_n: Nozzle exit area.
- D: Aerodynamic drag force acting against the rocket's motion $= \frac{1}{2}\rho V^2 C_d A_{ref}$

- $g(r)$: Gravitational acceleration as a function of altitude r.
- θ: Thrust angle relative to the vehicle axis.
- γ: Flight path angle.
- δ: Gimbal angle of the nozzle, representing the deviation of the thrust vector.

The negative terms are the losses, that in this computation are = 1775 m/s.

6.4 Second iteration

To optimize more, I took such number and I re-computed all the equations expressed so far in order to have a better estimation of the rocket. By doing so, I got new characteristics of the first stage that could be used again in the optimization:

1. Propellant mass: 375523 kg
2. Dry mass: 19151 kg
3. Specific impulse in vacuum: 311.6 s
4. Mass flow rate (9 engines): 2428.74 kg/s
5. Reference area (section area of the rocket): 9.6049 m^2
6. Nozzle exit area (9 engines): 6.8571 m^2

Thanks to these new numbers, I got a better optimization and estimation of the rocket. The new payload mass computed by the optimization is = 23.16 t (close to 23 t than before), and the new losses are: 1779 m/s. I stopped at this iteration.

A visualization of the orbit thanks to Google Earth can be seen in figure 1 and 2.

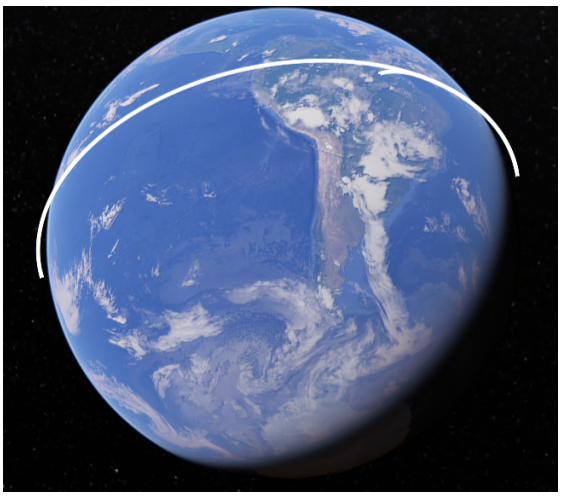

Figure 1:

Figure 2:

Let's now plot the main variables. Regarding the **altitude**, as we can also see in 3, we reached a maximum altitude of 249.037 km, that it's inside the 1-km from 250km constraint. Concerning the **velocity** (4), a maximum of 7276.76 m/s is reached while the V_{rel} required is 7277.52 m/s.

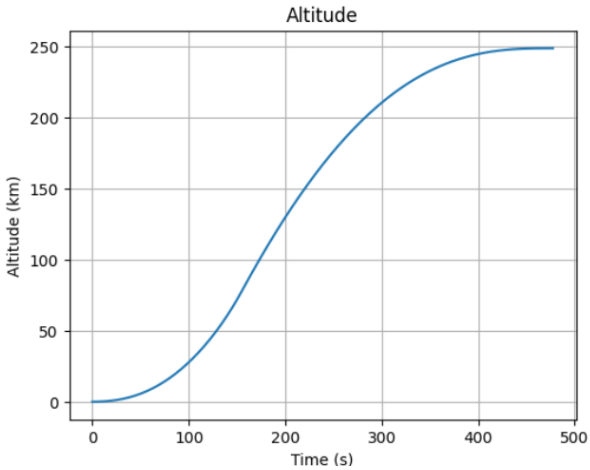

Figure 3:

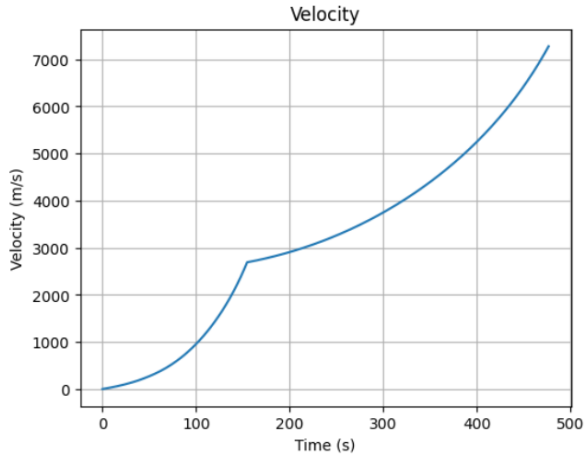

Figure 4:

The discrepancy is 0.756 m/s, that is between the 1-m/s constraint. In such graph, as in the others following, the effect of stage separation is clearly visible.

The **axial load factor** during the falcon 9 launch varies throughout the flight and is influenced by the rocket's thrust, drag, gravity, and mass distribution.
During the first stage, at liftoff, the thrust-to-weight ratio is typically just above 1 to ensure the rocket can lift off. As a result, the axial load factor (n_{axial}) starts close to 1 (1 g) because the thrust slightly exceeds the gravitational force (T/W = 1.35). As the rocket ascends and fuel burns, its mass decreases, causing the axial load factor to increase. In this phase, drag forces are significant due to the dense lower atmosphere, which reduces the net axial load factor slightly. The axial load factor peaks just before the first-stage engine cutoff due to reduced mass from fuel consumption and maximum thrust output.
During stage separation, there is a brief drop in the axial load factor as the first stage is jettisoned and the second-stage engine has not yet fully ignited.
For the second stage, the engine ignition occurs with a lower thrust-to-weight ratio compared to the first stage, causing the axial load factor to increase gradually. Similar to the first stage, the axial load factor peaks again near the second-stage engine cutoff due to minimized mass and constant thrust output.
This behavior can indeed be seen in figure 5.

Dynamic pressure (q in Pa) is given by:

$$q = \frac{1}{2}\rho V^2 \tag{46}$$

It starts at zero at liftoff, increases with velocity and air density, and peaks at max-q (figure 6). After max-q, as air density decreases with altitude, q sharply declines, eventually reaching zero in space. Max-q represents the point of maximum aerodynamic stress, typically occurring within the first minute of flight at altitudes of 10–20 km. Rockets usually throttle down near max-q to minimize loads, and ensure structural integrity and stability during ascent.

The **Flight Path Angle** (figure 7) represents the angle between the rocket's velocity vector and the local horizontal plane during its ascent. At the start of the launch, the flight path angle is steep ($> 80°$) as the rocket initially moves nearly vertically to overcome gravity and minimize atmospheric drag. As the Falcon 9 gains altitude and begins to accelerate horizontally to achieve orbital velocity, the flight path angle decreases steadily, approaching 0° at the end of the ascent when the trajectory becomes almost horizontal. This transition is essential for ensuring a smooth insertion into orbit. The minimum flight path angle is 0.0098°, that is less than the 0.01° constraint.

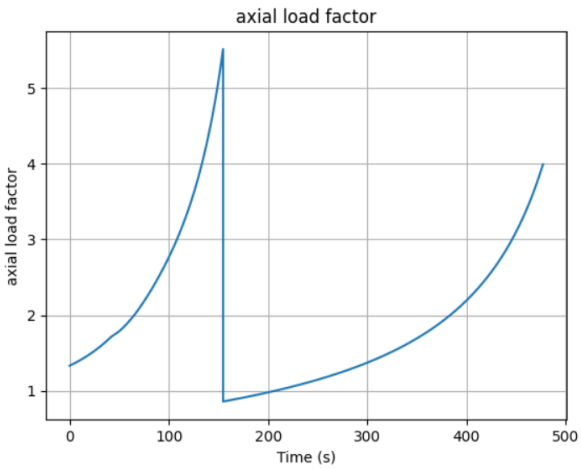

Figure 5: Axial Load Factor

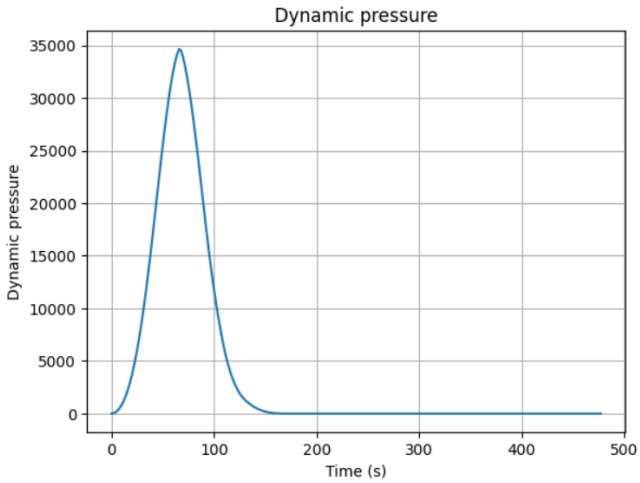

Figure 6: Dynamic Pressure

The **Pitch Angle** (figure 8) shows the typical gravity turn maneuver path, with an initial steep ascent before a gradual pitch-over to achieve the desired orbit.

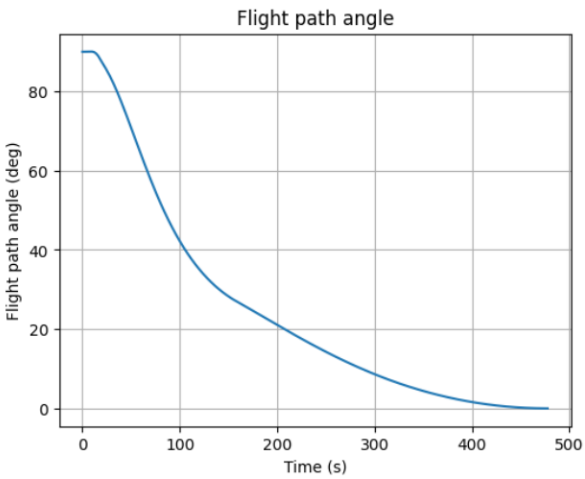

Figure 7: Flight Path Angle

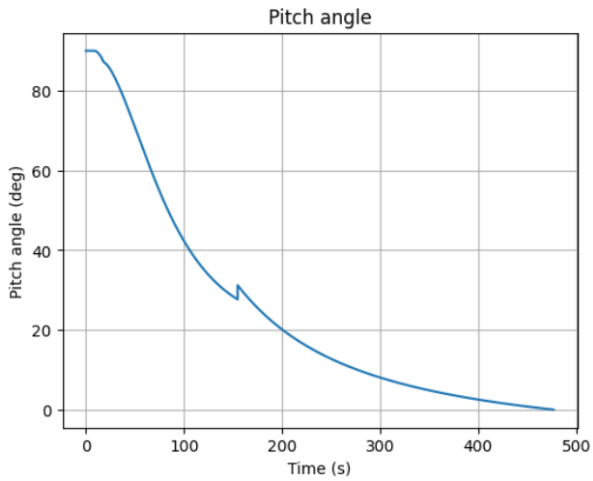

Figure 8: Pitch Angle

In conclusion, for the altitude, velocity, and flight path angle, we have verified that the constraints imposed for the last time step of the second stage (t_f) are satisfied.

6.5 Bonus Part

6.5.1 Position vector in ECI frame

In the simulation, the payload's position is given in spherical coordinates, defined by the radius r, longitude λ, and latitude δ. These need to be converted into cartesian coordinates using the following relations:

$$x = R\cos(\delta)\cos(\lambda), \quad y = R\cos(\delta)\sin(\lambda), \quad z = R\sin(\delta). \tag{47}$$

This transformation gives us the position vector in the Earth-Centered Earth-Fixed (ECEF) reference frame:

$$\mathbf{r}_{\text{ECEF}} = \begin{bmatrix} x_E \\ y_E \\ z_E \end{bmatrix}. \tag{48}$$

To express the position in the Earth-Centered Inertial (ECI) frame, we apply the following transformation matrix:

$$R_{\text{ECI}\to\text{ECEF}} = \begin{bmatrix} \cos\theta_E & \sin\theta_E & 0 \\ -\sin\theta_E & \cos\theta_E & 0 \\ 0 & 0 & 1 \end{bmatrix}. \tag{49}$$

Since this matrix transforms from ECI to ECEF, we take its transpose to convert the position from ECEF to ECI. The resulting position vector in the ECI frame is:

$$\mathbf{r}_{\text{ECI}} = R^{\top}_{\text{ECI}\to\text{ECEF}}\mathbf{r}_{\text{ECEF}} = \begin{bmatrix} x_{\text{ECI}} \\ y_{\text{ECI}} \\ z_{\text{ECI}} \end{bmatrix}. \tag{50}$$

6.5.2 Velocity vector in ECI frame

To determine the velocity in the ECI frame, a series of transformations are necessary. The velocity vector, as provided by the simulation, is initially expressed in the Aerodynamic frame. It is represented as:

$$\mathbf{v}_{\text{aero}} = \begin{bmatrix} \dot{x}_a \\ \dot{y}_a \\ \dot{z}_a \end{bmatrix} \tag{51}$$

The first transformation converts the velocity from the Aerodynamic frame to the Body frame. The transformation matrix is defined as:

$$R_{\text{aero}\to\text{body}} = \begin{bmatrix} \cos\alpha\cos\beta & -\cos\alpha\sin\beta & -\sin\alpha \\ \sin\beta & \cos\beta & 0 \\ \sin\alpha\cos\beta & -\sin\alpha\sin\beta & \cos\alpha \end{bmatrix} \tag{52}$$

The velocity in the Body frame is then calculated as:

$$\mathbf{v}_{\text{body}} = R_{\text{aero}\to\text{body}} \cdot \mathbf{v}_{\text{aero}} \tag{53}$$

Next, the velocity is transformed from the Body frame to the Local Tangent Plane (LTP) frame using the matrix:

$$R_{\text{body}\to\text{LTP}} = \begin{bmatrix} \cos\psi\cos\theta & \sin\psi\cos\theta & -\sin\theta \\ \cos\psi\sin\phi\sin\theta - \sin\psi\cos\phi & \sin\psi\sin\phi\sin\theta + \cos\psi\cos\phi & \cos\theta\sin\phi \\ \cos\psi\sin\phi\cos\theta + \sin\psi\sin\phi & \sin\psi\sin\phi\cos\theta - \cos\psi\sin\phi & \cos\theta\cos\phi \end{bmatrix} \tag{54}$$

The velocity in the LTP frame is computed as:

$$\mathbf{v}_{\text{LTP}} = R_{\text{body}\to\text{LTP}} \cdot \mathbf{v}_{\text{body}} \tag{55}$$

From the LTP frame, the velocity is transformed to the ECEF frame using the matrix:

$$R_{\text{ECEF}\to\text{LTP}} = \begin{bmatrix} -\cos\lambda_g\sin\phi_g & -\sin\lambda_g\sin\phi_g & \cos\phi_g \\ -\sin\lambda_g & \cos\lambda_g & 0 \\ -\cos\lambda_g\cos\phi_g & -\sin\lambda_g\cos\phi_g & -\sin\phi_g \end{bmatrix} \tag{56}$$

The velocity in the ECEF frame is given by:

$$\mathbf{v}_{\text{ECEF}} = R^{\text{T}}_{\text{ECEF}\to\text{LTP}} \cdot \mathbf{v}_{\text{LTP}} \tag{57}$$

Finally, to obtain the velocity in the ECI frame, we use the transformation matrix from ECEF to ECI:

$$\mathbf{v}_{\text{ECI}} = R^{\text{T}}_{\text{ECEF}\to\text{ECI}} \cdot \mathbf{v}_{\text{ECEF}} \tag{58}$$

6.5.3 Verification of Orbit

From the velocity $\mathbf{v}$ and position $\mathbf{r}$ in the Earth Centered Inertial (ECI) frame, the eccentricity vector can be defined as:

$$\mathbf{e} = \frac{1}{\mu}\left[\left(v^2 - \frac{\mu}{r}\right)\mathbf{r} - (\mathbf{r}\cdot\mathbf{v})\mathbf{v}\right] \tag{59}$$

where v is the norm of the velocity vector in the ECI frame, r is the norm of the position vector in the ECI frame, and μ is the standard gravitational parameter.
The energy of the orbit is given by:

$$E = \frac{v^2}{2} - \frac{\mu}{r} \tag{60}$$

The semi-major axis of the orbit is expressed as:

$$a = -\frac{\mu}{2E} \tag{61}$$

From these values we can verify that the apogee altitude and eccentricity of the orbit correspond to what we obtained before. However, for time reasons, I was not able to verify it numerically.

Conclusion

This report provides a comprehensive design and optimization framework for a rocket launcher based on the Falcon 9 architecture. Each step, from propulsion to trajectory optimization, was analyzed to ensure the mission's success.
The following key outcomes were achieved:

- **Payload Mass**: Successfully optimized to 23.16 tons, exceeding the initial requirement of 23 tons.
- Δv **Requirements**: Total velocity increment calculated as $\Delta v_{\text{req}} = 9095.706\,\text{m/s}$, with losses of 1779 m/s after optimization.
- **Stage Masses**:
  - First Stage: Propellant mass 375523 kg, dry mass 19151 kg.
  - Second Stage: Propellant mass 92670 kg, dry mass 3900 kg.
- **Geometrical Dimensions**:
  - Total Rocket Length: 65.71 m.
  - Diameter: 3.514 m.
- **Performance Metrics**:
  - Maximum Altitude: 249.037 km, within the 1 km constraint.
  - Maximum Velocity: 7276.76 m/s, deviating by only 0.756 m/s from the required V_{rel}.
  - Flight Path Angle: Minimum of 0.0098°, satisfying the 0.01° constraint.

These results validate the proposed rocket design, showcasing its capability to achieve the required mission objectives while meeting the imposed constraints.